

The GitOps controller keeps checking the deployment repository all the time. Once it sees the approved configuration has changed, it synchronises the target Kubernetes environment with the new desired state. The model can be released in a controlled way – for example, canary deployment with only a part of the requests sent to the new model.

After being deployed, the model is monitored with respect to prediction quality, delay, resource usage, data drift, and model drift. If any critical problem arises, the earlier stable version can be reverted via a Git rollback.

Version control and model lifecycle management

Version control is a crucial element of the entire lifecycle of an artificial intelligence model. An AI system features various changing elements, including source code, datasets, preprocessing, model parameters, data obtained because of training, and configuration files that were used for deployment. If these are not stored, the process of recreating the process of experimentation or finding out which version of the model is currently being used becomes almost impossible.

Git manages source code, pipeline definitions and configuration files, and data version control is used for controlling massive datasets and model files. For example, MLflow allows to keep a record of the parameters of the model, evaluation metrics, and results of the experiments.

Successful models can be recorded in the model registry once they are trained. Models that are recorded in this system can be considered belonging to a certain stage of the lifecycle (such as development, validation or staging). A model can be included in production only if it provided positive results during the tests.

When the production version undergoes a declining performance level, uncontrollable delays, or model shift, the team can determine the specific version of the product in question and access its older versions. Once this is successful, it becomes easy to get back to the appropriate model by returning the needed Git settings. This version-controlled lifecycle provides improved reproducibility, management, and responsibility, and includes an accurate log of how each model was made, evaluated, approved, implemented, and substituted.

Open source GitOps ecosystem

The GitOps open source community consists of tools that can assist in various tasks throughout the lifecycle of artificial intelligence — from source code management to infrastructure provision of models. Organisations no longer need to rely solely on a single platform to do their work. They can combine as many tools as their technical needs require.

Git repositories are the basis for this process because they contain the code, pipelines, and deployments in a version-controlled format. DVC expands upon this by keeping track of huge datasets and any other items that can't be stored inside Git. MLflow, on the other hand, is designed for experiment tracking and model management. Kubeflow allows teams to create scalable machine learning pipelines in Kubernetes. It can perform tasks such as preprocessing, training, validation, and serving of models automatically. Argo CD and Flux serve as GitOps controllers. They are constantly comparing configurations stored inside Git and applied to the Kubernetes environment. Terraform and OpenTofu allow organisations to practice Infrastructure as Code (IaC) by providing cloud resources, networks,

Comparison of open source tools for AI GitOps

Tool	Category	Primary purpose	Role in AI workloads	Key strength
Git	Version control	Tracks code and configuration changes	Maintains pipeline and deployment definitions	Complete change history
DVC	Data versioning	Tracks datasets and large model files	Connects data versions with experiments	Reproducible datasets
MLflow	ML lifecycle management	Tracks experiments and model versions	Records metrics, parameters and model stages	Central model registry
Kubeflow	ML pipeline orchestration	Builds and executes ML workflows	Automates training, evaluation and serving	Kubernetes-native scalability
Argo CD	GitOps deployment	Synchronises Git with Kubernetes	Deploys approved model configurations	Continuous reconciliation
Flux	GitOps deployment	Automates Kubernetes configuration updates	Maintains the desired deployment state	Lightweight automation
Terraform/ OpenTofu	Infrastructure as Code	Provisions computing infrastructure	Creates clusters, storage and GPU resources	Repeatable infrastructure